Theoretical Foundations of Quantum Error-Correction

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Contents

Abstract **v**

1 Error Model **1**

2 Classical and Quantum Coding Theory **7**

Classical Codes 7

Quantum Codes 7

Recovery Conditions 8

Knill-Laflamme Conditions 9

Characteristics Imposed on a Code by the Knill-Laflamme Conditions . . . 12

Degenerate Codes 12

Quantum Hamming Bound 13

3 Repetition Code **15**

4 Quantum Repetition Code **19**

5 Linear Codes **21**

Bibliography **24**

Abstract

Quantum computing (QC) is a promising paradigm that offers potentially exponential advantages over classical computing for specific problems, such as factorization, quantum simulation, and optimization. A key property enabling these benefits is the linearity of quantum mechanics, which, in principle, prevents errors from accumulating between gates (Kitaev, Shen and Vyalyj, 2002). However, the inherent quantum nature and extreme sensitivity of these systems to errors, which can be introduced by gates or external conditions (e.g., environmental fluctuations, electromagnetic interference), mean that linearity alone is insufficient to ensure reliable results, particularly for large-scale operations like factorization (Calderbank and Shor, 1996). Furthermore, the no-cloning theorem prohibits the copying of qubits, rendering traditional error correction methods, such as redundancy, inapplicable.

Quantum error-correction (QEC) is a field developed to address this challenge. It was demonstrated that quantum information could be encoded non-locally by distributing it across multiple qubits (Calderbank and Shor, 1996). Codes that employ this strategy are known as quantum error-correcting codes (QECCs). The efficacy of this method has been supported by experimental improvements in physical quantum computers (AbuGhanem, 2025). Nevertheless, quantum error-correction remains an open area of research, and advancements in QEC are critical for the continued development of QC (AbuGhanem, 2025).

Error Model

Creating a general mathematical model for error correction means that we should make as few assumptions as necessary for the error (or noise) model that we will work with. That way, if we define a method for handling these general errors, it can be specialized to any other possible error. However, for simplicity, we will still make some assumptions. For instance, we will work in a two-dimensional Hilbert space for every qubit (this is not required, but it simplifies the examples and conclusions; when necessary, explanations for generalizations will be provided). We also make the broad assumption that the error is defined by an operation on the tensor product space of the qubit space and an environment space. To this end, suppose we have a qubit in the state $|\psi\rangle = \alpha_0 |0\rangle + \alpha_1 |1\rangle$ and an environment in the state $|E\rangle$. We define the error as the operation

$$|\psi\rangle |E\rangle \;\rightarrow\; \alpha_0\beta_1 |0\rangle |E_1\rangle + \alpha_0\beta_2 |1\rangle |E_2\rangle + \alpha_1\beta_3 |0\rangle |E_3\rangle + \alpha_1\beta_4 |1\rangle |E_4\rangle \,.$$

Thus, for a general qubit, an error is an interaction that, with certain amplitudes, produces a new environment state (which could be different for each possible environment basis state) and also changes the state of the qubit with some probability. At first glance, this seems unreasonable because we would be dealing with an uncountably infinite number of possible errors. However, by the theorem [](pauli is a base) we know that the set of operations $\mathcal{B} = \{I, Z, X, XZ\}$ forms a basis on a two-dimensional Hilbert space. This implies that every possible error can be rewritten as a linear combination of that set. Furthermore, we can

simplify our previous expression to reflect this fact (Kaye, Laflamme and Mosca (2006)).

$$
\begin{aligned}
&\alpha_0\beta_1 |0\rangle |E_1\rangle + \alpha_0\beta_2 |1\rangle |E_2\rangle \\
&+\alpha_1\beta_3 |0\rangle |E_3\rangle + \alpha_1\beta_4 |1\rangle |E_4\rangle \\
&= \frac{1}{2} \left((\alpha_0 |0\rangle + \alpha_1 |1\rangle) (\beta_1 |E_1\rangle + \beta_3 |E_3\rangle) \right. \\
&\quad + (\alpha_0 |0\rangle - \alpha_1 |1\rangle) (\beta_1 |E_1\rangle - \beta_3 |E_3\rangle) \\
&\quad + (\alpha_0 |1\rangle + \alpha_1 |0\rangle) (\beta_2 |E_2\rangle + \beta_4 |E_4\rangle) \\
&\quad \left. + (\alpha_0 |1\rangle - \alpha_1 |0\rangle) (\beta_2 |E_2\rangle - \beta_3 |E_4\rangle) \right) \\
&= \frac{1}{2} \left(I |\psi\rangle (\beta_1 |E_1\rangle + \beta_3 |E_3\rangle) \right. \\
&\quad + Z |\psi\rangle (\beta_1 |E_1\rangle - \beta_3 |E_3\rangle) \\
&\quad + X |\psi\rangle (\beta_2 |E_2\rangle + \beta_4 |E_4\rangle) \\
&\quad \left. + ZX |\psi\rangle (\beta_2 |E_2\rangle - \beta_3 |E_4\rangle) \right)
\end{aligned}
$$

One important detail is that we have not expressed the new environment states in terms of anything else because we do not know the structure nor the specifications of the environment space. Moreover, this result is only valid for a 2 dimentional Hilbert space as explained before. However, the underlaying result, of being able to divide every continuos error into a basis of errors, would hold for any dimention.

Generalizing this model for an state that live into a more than one qubit is done by generalizing the error into an unitary operation U_{err} that affects the joint space. This means that:

1 You can describe it as a linear combination of operations that affect the state of the qubit and the state of the environment.

2 You can represent the matrix of the hole operation with the matrix given by

$$\rho = U_{err} |\psi\rangle |E\rangle \langle E| \langle\psi| U_{err}^{\dagger} \tag{1.1}$$

3 You can specialize the previous result to get the particular state by calculating it's trace.

$$Tr(\rho) = Tr(U_{err} \left|\psi\right\rangle \left|E\right\rangle \left\langle E\right| \left\langle\psi\right| U_{err}^{\dagger}) = \sum_i \varepsilon_i^Q \left|\psi\right\rangle \left\langle\psi\right| \varepsilon_i^{Q\dagger} \tag{1.2}$$

4 The error can be express as the applyance of the new operators ε_i^Q witch encodes all the information necessary including the environment.(Kaye, Laflamme and Mosca (2006))

Theorem 1.1. *The set of Pauli operations $\mathcal{B} = \{I, Z, X, XZ\}$ forms a basis for the operators on a two-dimensional Hilbert space.*

Proof.
Because the cardinality of $\mathcal{B}$ equals the dimension of the space of operators on the Hilbert space, it is sufficient to show that the set is linearly independent. We use the trace inner product (see Definition [](Trace Class Norm)), which defines an inner product on 2×2 matrices.

$$
\begin{aligned}
\langle I, X\rangle &= Tr\left(\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}^* \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\right) = Tr\left(\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\right)\\
&= 0\\
\langle I, Z\rangle &= Tr\left(\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}^* \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\right) = Tr\left(\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\right)\\
&= 0\\
\langle I, XZ\rangle &= Tr\left(\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}^* \begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}\right) = Tr\left(\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}\right)\\
&= 0\\
\langle X, Z\rangle &= Tr\left(\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}^* \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\right) = Tr\left(\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}\right)\\
&= 0\\
\langle X, XZ\rangle &= Tr\left(\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}^* \begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}\right) = Tr\left(\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\right)\\
&= 0\\
\langle Z, XZ\rangle &= Tr\left(\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}^* \begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}\right) = Tr\left(\begin{pmatrix}0 & -1\\ -1 & 0\end{pmatrix}\right)\\
&= 0
\end{aligned}
$$

All pairwise inner products are zero, so the set is orthogonal and therefore linearly independent. By Theorem [](Theorem of Orthogonality to Lineality), every unitary operator on a Hilbert space can be represented as a linear combination of the elements of $\mathcal{B}$.

Definition 1.2. For $n \times n$ matrices, we define the *trace class norm* (inner product) as

$$\langle A, B\rangle = Tr(A^*B). \tag{1.3}$$

Checks.
We verify the inner-product properties:

1 Conjugate symmetry:

$$\overline{\langle B, A\rangle} = \overline{Tr(B^*A)} = Tr((B^*A)^*) = Tr(A^*B) = \langle A, B\rangle.$$

2 Linearity in the second argument:

$$\begin{aligned}\langle A, \alpha B + \beta C\rangle &= Tr\big(A^*(\alpha B + \beta C)\big)\\ &= Tr(\alpha A^*B + \beta A^*C)\\ &= \alpha Tr(A^*B) + \beta Tr(A^*C)\\ &= \alpha\langle A, B\rangle + \beta\langle A, C\rangle.\end{aligned}$$

3 Positive definiteness:

$$\langle A, A\rangle = Tr(A^*A) = \sum_{i=1}^{n}(A^*A)_{ii} = \sum_{j=1}^{n}\sum_{i=1}^{n}\overline{a}_{ji}a_{ji}.$$

For every complex number x, $\overline{x}x \geq 0$, so $\langle A, A\rangle \geq 0$. Moreover, $\overline{x}x = 0$ iff $x = 0$; therefore $\langle A, A\rangle = 0$ implies $A = 0$.

Theorem 1.3. *Given a set $\mathcal{B}$ of non-zero vectors in a vector space of dimension n and an inner product $\langle\cdot,\cdot\rangle$,*

$$\forall v_1, v_2 \in \mathcal{B},\ v_1 \neq v_2 :\ \langle v_1, v_2\rangle = 0 \implies \mathcal{B} \text{ is linearly independent.} \tag{1.4}$$

Proof.
Let $\sum_{i=1}^{n} a_i v_i = 0$. Take an arbitrary v_k and the inner product of both sides with v_k:

$$\left\langle v_k,\ \sum_{i=1}^{n} a_i v_i\right\rangle = \langle v_k, 0\rangle. \tag{1.5}$$

Because $\mathcal{B}$ is orthogonal, $\langle v_k, v_i\rangle = 0$ for all $i \neq k$; hence every term except the k-th vanishes:

$$a_k\langle v_k, v_k\rangle = 0. \tag{1.6}$$

Since $v_k \neq 0$, we have $\langle v_k, v_k\rangle > 0$, thus $a_k = 0$. Repeating this for each k shows that all coefficients are zero, so $\mathcal{B}$ is linearly independent.

Classical and Quantum Coding Theory

Classical Codes

In classical information theory, a code is defined as a subset $M \subseteq \mathbb{B}^n$ containing 2^m elements (Kitaev, Shen and Vyalyj (2002)). Notably, m is not restricted to integer values; for a code with six possible values, $m = \log_2(6) \approx 2.5849$. The difference between the physical block length n and the message length m defines the redundancy of the system. This redundancy is a fundamental requirement for verifying the integrity of a received bitstring. While these subsets typically map to external data structures, such as the ASCII standard where $m = 7$, they more broadly function as the essential interface between physical bit states and logical information.

In the context of error correction, codes are structured such that the underlying information remains recoverable despite noise. Classical errors are typically discrete, manifesting as "bit-flips" within a string. Consequently, techniques such as data redundancy and the cloning of information are central to classical error mitigation. It is important to note that these specific strategies are prohibited in the quantum regime due to the No-Cloning Theorem.

Quantum Codes

In quantum computation, defining a code as a discrete subset of bitstrings is insufficient, as it fails to account for the continuous nature of quantum states.

Instead, a quantum code is defined as a subspace of the total Hilbert space (Kitaev, Shen and Vyalyj (2002)). This formulation allows for the encoding of arbitrary superpositions of basis states. By leveraging the linearity of vector spaces, one can define an encoding transformation by mapping the original basis vectors into a linearly independent set that spans the protected code subspace.

This subspace-based approach ensures that linear transformations can be applied while the state remains within the code's boundaries. Furthermore, it provides the mathematical framework necessary to identify and rectify errors without collapsing the logical state. This is achieved by extracting an error syndrome, a diagnostic indicator of the noise process, via projective measurements onto the subspace, rather than by measuring individual physical qubits.

Recovery Conditions

To maintain generality in the theory of quantum error-correcting codes, minimal assumptions are made regarding the noise model. We consider an error channel $\mathcal{E}$ acting on a density matrix ρ as a trace-preserving map:

$$\mathcal{E}(\rho) = \sum_i E_i \rho E_i^\dagger \tag{2.1}$$

where $\{E_i\}$ are the Kraus operators representing the noise process. A code C is said to correct the channel $\mathcal{E}$ if there exists a recovery operation $\mathcal{R}$ such that (Nielsen and Chuang (2010)):

$$\forall \rho \in C \implies (\mathcal{R} \circ \mathcal{E})(\rho) \propto \rho \tag{2.2}$$

The map $\mathcal{R}$ is a physical process designed to map the corrupted state back into the code space. Critically, the recovery is considered successful if the logical information is preserved, even if the resulting physical state is not identical to the original codeword. This distinction is vital for understanding degenerate codes, where multiple distinct error processes may lead to the same corrected state within the logical subspace.

Knill-Laflamme Conditions

The existence of a recovery operation for an error is not guaranteed for an arbitrary code. Therefore, it is crucial to have a method to determine if a code can correct an error $\mathcal{E}$ within the subspace spanned by $\{E_i\}$. For a code $\mathcal{C}$ with projector P, the necessary and sufficient condition for the code to correct the error $\mathcal{E}$ is given by the Knill-Laflamme conditions (Nielsen and Chuang (2010)):

$$PE_i^\dagger E_j P = \alpha_{ij} P \tag{2.3}$$

for some Hermitian matrix α.

Proof (adapted from Nielsen and Chuang (Nielsen and Chuang, 2010)):

To prove sufficiency, we can construct the operation $\mathcal{R}$ assuming the Knill-Laflamme conditions are fulfilled. We assume a general method for error correction that separates syndrome measurement and error recovery, proving that both components can be generated given these conditions.

By assumption, α is a Hermitian matrix and is therefore diagonalizable as $d = u^\dagger \alpha u$. We can define operations $F_k := \sum_i u_{ik} E_i$, and given Theorem 2.1, $\{F_k\}$ is also a set of operation elements for $\mathcal{E}$. By substitution, we get:

$$\begin{aligned}
PF_k^\dagger F_l P &= \sum_{ij} u_{ki}^\dagger u_{jl} P E_i^\dagger E_j P \\
&= \sum_{ij} u_{ki}^\dagger u_{jl} \alpha_{ij} P \\
&= \sum_{ij} u_{ki}^\dagger \alpha_{ij} u_{jl} P \\
&= d_{kl} P
\end{aligned}$$

Using the polar decomposition for $F_k P$, we obtain:

$$
\begin{aligned}
F_k P &= U_k \sqrt{(F_k P)^\dagger F_k P} \\
&= U_k \sqrt{P F_k^\dagger F_k P} \\
&= U_k \sqrt{d_{kk} P} \\
&= \sqrt{d_{kk}} U_k P
\end{aligned}
$$

Therefore, F_k rotates into a subspace projected by $P_k := U_k P U_k^\dagger = \frac{F_k P U_k^\dagger}{\sqrt{d_{kk}}}$. Moreover, given that d is a diagonal matrix, it is evident that the spaces projected by P_k are orthogonal, allowing for a reliable syndrome measurement. This could also be augmented into a trace-preserving operation by adding the necessary subspaces (though this is not crucial in every code, such as the repetition code). The error correction step would simply involve applying $U_k^\dagger$.

Thus, we can construct the operation $\mathcal{R}(\sigma) = \sum_k U_k^\dagger P_k \sigma P_k U_k$. We can show that it satisfies the recovery condition because, without loss of generality for $\rho \in \mathcal{C}$:

$$
\begin{aligned}
\mathcal{R}(\mathcal{E}(\rho)) &= \sum_{kl} U_k^\dagger P_k F_l \rho F_l^\dagger P_k U_k \\
&= \sum_{kl} \left(U_k^\dagger P_k F_l \sqrt{\rho} \right) \left(\sqrt{\rho} F_l^\dagger P_k U_k \right) \\
&= \sum_{kl} \left(U_k^\dagger P_k F_l \sqrt{\rho} \right) \left(U_k^\dagger P_k F_l \sqrt{\rho} \right)^\dagger
\end{aligned}
$$

Evaluating the inner term:

$$
\begin{aligned}
U_k^\dagger P_k F_l \sqrt{\rho} &= U_k^\dagger P_k F_l P \sqrt{\rho} \\
&= \frac{U_k^\dagger U_k P F_k^\dagger F_l P \sqrt{\rho}}{\sqrt{d_{kk}}} \\
&= \frac{\delta_{kl} d_{kk} P \sqrt{\rho}}{\sqrt{d_{kk}}} \\
&= \delta_{kl} \sqrt{d_{kk}} \sqrt{\rho}
\end{aligned}
$$

Substituting this back yields:

$$\begin{aligned}\mathcal{R}(\mathcal{E}(\rho)) &= \sum_{kl} \left(\delta_{kl}\sqrt{d_{kk}}\sqrt{\rho}\right)\left(\delta_{kl}\sqrt{d_{kk}}\sqrt{\rho}\right)^{\dagger} \\ &= \sum_{kl} \delta_{kl} d_{kk} \rho \\ &\propto \rho\end{aligned}$$

This satisfies the conditions for C to correct $\mathcal{E}$.

To prove the necessity of the Knill-Laflamme conditions, suppose $\mathcal{E}$ is perfectly correctable by the recovery operation $\mathcal{R}$ with operation elements $\{R_j\}$. Define an operation $\mathcal{E}_C(\rho) = \mathcal{E}(P\rho P)$ which is also recoverable by $\mathcal{R}$, where the proportionality factor is a constant c independent of ρ. This can be rewritten as:

$$\begin{aligned}\mathcal{R}(\mathcal{E}_C(\rho)) &= cP\rho P \\ \sum_{ij} R_j E_i P \rho P E_i^{\dagger} R_j^{\dagger} &= cP\rho P\end{aligned}$$

By Theorem 2.1, we know there exist complex numbers in a unitary matrix c such that:

$$\begin{aligned}R_k E_i P &= c_{ki} P \\ P E_i^{\dagger} R_k^{\dagger} &= c_{ki}^{*} P \\ P E_i^{\dagger} R_k^{\dagger} R_k E_j P &= c_{ki}^{*} c_{kj} P\end{aligned}$$

Considering that $\mathcal{R}$ is trace-preserving, summing over k yields:

$$P E_i^{\dagger} E_j P = \sum_k c_{ki}^{*} c_{kj} P = \alpha_{ij} P \tag{2.4}$$

This perfectly matches the Knill-Laflamme conditions.

Characteristics Imposed on a Code by the Knill-Laflamme Conditions

The Knill-Laflamme conditions are a great way to reveal that, due to the existence of α as a Hermitian matrix and the conditions themselves, a quantum code that corrects an error needs to be:

1. **Orthogonal:** The basis should be orthogonal so that the states are distinguishable, which is reflected by the syndrome measurement operation obtained in the proof of the Knill-Laflamme conditions. This is also indispensable for α to be a Hermitian matrix (Kitaev, Shen and Vyalyj (2002)).

2. **Non-deformed by errors:** This corresponds, by the definition of deformation, to the Knill-Laflamme conditions. It basically represents that a codeword should not be changed or made non-orthogonal by the application of the error. Thus, even when the error is applied, it is still possible to distinguish between states and recover them.

Lemma 2.1 (Theorem 8.2, Nielsen and Chuang (Nielsen and Chuang, 2010))**.** *Suppose $\{E_1, E_2, \ldots, E_m\}$ and $\{F_1, F_2, \ldots, F_n\}$ are operation elements giving rise to quantum operations $\mathcal{E}$ and $\mathcal{F}$ respectively. By appending zero operators to the shorter list, we may ensure that $m = n$. Then $\mathcal{E} = \mathcal{F}$ if and only if there exist complex numbers u_{ij} such that $E_i = \sum_j u_{ij} F_j$, where u is an $m \times m$ unitary matrix.*

Degenerate Codes

As seen in the previos sections the classical and quantum codes are considerably intertwined. However, degenrate codes are a family of quantum codes without any corresponding classical system. In classical computing the application of errors in different bits necessary lead into diferent codewords. However, in quantum codes an error applied in different qubits can produce the same codeword.

Lemma 2.2 (Degenerate codes have a singular hermitian matrix for the Knill-Laflamme Conditions)**.** *TODO: Debo demostrar esto*

Examples of this types of codes would come later and would be a crutial characteristic because even thou the exact error would not be distinguishable the correction of it can be done by the same method. Nonetheless, it's important to consider that this distinction would also carry the cost of not being able to apply some of the methods for property demonstration would not be applicable to this subset of codes. One relevant example would come in the next section about quantum hamming bound.

Quantum Hamming Bound

Given a set of errors there are multiple codes that can correct it. Therefore for any application it can be look for a code that minimize the size of the logical qubit. The Quantum Hamming Bound is way to find the lower bound for a non-degenerate quantum code.

Definition 2.3 (Quantum Hamming Bound)**.** Let C be a non-degenerate code from a space of dimention k into a subspace of dimention n that can correct arbitrary-errors in at most t qubits. The smallest logical qubit for C should hold that

$$\sum_{j=0}^{t} \binom{n}{j} 3^j 2^k \leq 2^n \tag{2.5}$$

Deduction:

For any subset of errors happening in $j \leq t$ qubits the possible combinations of errors are the binomial for selecting j qubits in the n bitstring $\binom{n}{j}$. For every qubit in that sistem the error that could happend would be of dimention 4 (as demonstrated in the theorem 1.1). However, for every qubit the case for the operation I would be discarded as applying it would not change the result in any way whatsoever. Therefore for every set of positions there is 3^j possibilities of errors. By assumption C is non-degenerate and therefore every combination of errors would need to correspond into a 2^k-space. Since C corrects every possible $j \leq t$ the smallest logical qubit for C should hold that

This bound creates a rule of thumb for minimizing the size of the logical qubit in a Code. However, as a whole the lack of consideration for degenerate codes could be an important caviat for the system.

(OPTIONAL) As a side not is important to mention that in

Example 2.4 (Hamming Bound for Code that corrects arbitrary errors in a single qubit)**.** Let C be an error correcting code that can correct any error in single qubit. Applying the Hamming Bound in it we get

$$\begin{aligned}
\sum_{j=0}^{1} \binom{n}{j} 3^j 2^k &\leq 2^n \\
\left[\binom{n}{0} 3^0 2^1 + \binom{n}{1} 3^1 2^1\right] &\leq 2^n \\
[2 + n \cdot 3 \cdot 2] &\leq 2^n \\
2\,[1 + 3n] &\leq 2^n
\end{aligned}$$

It is apparent here that n must be equal or greater than 5 for it to hold. Therefore the smallest non-degenerate quantum code that can correct arbitrary errors

Repetition Code

Suppose a noisy channel in which, with probability p, a bit that passes through flips. This implies that we will only get the correct information with probability $(1-p)$. We can obtain better results by encoding the information. In this case, suppose we are taking a single bit of information and encoding it with three additional bits. A particular aspect of this model to consider is that every bit in that sequence holds the same amount of information. This is a strictly classical concept called *redundancy*.

$$1 \longrightarrow 111$$

$$0 \longrightarrow 000$$

Figure 3.1: Representation of the repetition code. This image was inspired by the book (Nielsen and Chuang, 2010). It shows that each bit is represented by three bits—a strictly classical form of redundancy.

After passing through the channel, we obtain a new bit string. For the sake of example, let's call it S_n. This bit string could be the original one or any other

combination of the three bits, depending on how many times an error occurred. Decoding is where we gain most of the benefits of using a code in the first place, and in this example there are two primary decoding methods to improve fidelity:

1. We could build a majority-vote system, meaning we interpret the bit string as representing bit 0 if the majority of elements are 0, and similarly for 1.

2. The other decoding method is to check if two bits are equal. This detects that an error occurred and indicates how to correct it. In this particular case, you can compare the first and second bits, and the first and third bits.

These two systems work essentially the same way in classical computing theory. However, the first method cannot be directly translated into quantum computing because it requires measuring each bit independently. Therefore, we will focus on the second option for the decoding process so that a parallel can be drawn with the quantum equivalent of this code.

For ease of understanding, consider the case where the decoding process receives the bit string 010. In that scenario, the decoding process would indicate that the first and second bits are different and that the first and third bits are equal, implying that the second bit was the source of the error.

However, that could be the wrong assumption; it is possible that the first and third bits were actually the ones affected by the error. This indistinguishability between the two possibilities indicates that this code works only if an error occurs in at most one bit. Therefore, the probability of an error with this code is

$$p^3 + 3p^2(1-p) = 3p^2 - 2p^3. \tag{3.1}$$

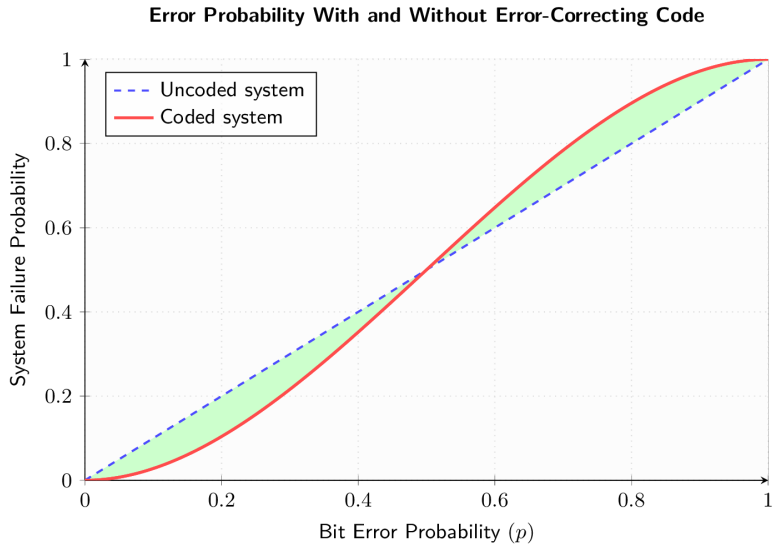

Figure 3.2: Comparison of the probability of error for a bit transmitted through the channel with and without the code. The dashed blue line represents the unprotected bit, whose error probability is linear. The red line represents the coded system, which performs better when the error probability is less than 50%.

Note that without coding, the probability of failure is p. Therefore, this code is only helpful if $3p^2 - 2p^3 \leq p$, which holds only when $p \leq \frac{1}{2}$. One can actually increase the number of bits used to encode the same amount of information and obtain different thresholds of usefulness—there has been extensive work in classical coding theory on this. However, the core idea of encoding information is already captured by this example.

Quantum Repetition Code

Given that we have already seen the classical repetition code in Section 3, we can now take the first leap into a quantum code. This code relies on essentially the same theoretical principle. However, the decoding process must be interpreted through a different lens because we cannot duplicate the exact same information across every qubit. Rather, the information is distributed among all three qubits.

The first defining characteristic of this code is its encoding process. In a classical code, a bit can simply be duplicated, which is impossible in quantum computing due to the no-cloning theorem. Nonetheless, we can construct a unitary operator that encodes a single logical qubit into three physical qubits. This is illustrated by the circuit in Figure 4.1.

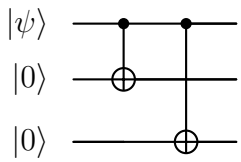

Figure 4.1: Circuit that encodes a single qubit in a manner equivalent to the classical repetition code. This demonstrates that encoding is possible due to the existence of a linear, Hermitian process.

Once encoded, the logical codespace is defined by the following projector:

$$P = |000\rangle\langle 000| + |111\rangle\langle 111|$$

The decoding process, as demonstrated previously, is divided into two stages. First, we obtain the error syndrome, which is achieved by applying the following projectors:

- $P_0 = |000\rangle\langle 000| + |111\rangle\langle 111|$ (No error occurred)
- $P_1 = |100\rangle\langle 100| + |011\rangle\langle 011|$ (Error in the first qubit)
- $P_2 = |010\rangle\langle 010| + |101\rangle\langle 101|$ (Error in the second qubit)
- $P_3 = |001\rangle\langle 001| + |110\rangle\langle 110|$ (Error in the third qubit)

Second, we correct the error by applying the appropriate inverse operation based on the syndrome obtained in the previous step. This approach is analogous to the classical problem; therefore, the analysis of fidelity is identical and yields equivalent results.

However, because this is a quantum code, we can explicitly demonstrate its ability to correct bit-flip errors using the Knill-Laflamme conditions. Given our previously defined projector and the set of single-qubit bit-flip error operators:

$$\{E_i\} = \{I \otimes I \otimes I, X \otimes I \otimes I, I \otimes X \otimes I, I \otimes I \otimes X\}$$

We can now show that the Knill-Laflamme conditions hold for this code. First, considering the cases where $i = j$ (since $E_i^\dagger E_i = I$ for Pauli matrices), the condition reduces to:

$$\begin{aligned} PE_i^\dagger E_i P &= PIP \\ &= P^2 \\ &= P \end{aligned}$$

Therefore, $\alpha_{ii} = 1$. Conversely, for $i \neq j$, applying the error operators to the projector P yields P_{ij}, which is the original operator modified by the application of E_i and $E_j^\dagger$. This resulting operator is orthogonal to the original projector, meaning $\alpha_{ij} = 0$. In summary, the matrix α is the identity matrix, which is Hermitian, as required.

Linear Codes

Definition 5.1. A code c that encodes k bits of information into an n-bit space is called *linear* if every possible string to be encoded can be represented as a k-dimensional binary vector v and it is encoded by the operation gv, where g is an $n \times k$ matrix with entries in $\mathbb{Z}_2$, independent of v, called a *generator matrix*. The vectors that form g are typically chosen as a basis for the subspace of codewords. Thus, every operation is performed modulo two and every possible codeword is a linear combination of these basis vectors. **?**

Theorem 5.2. *Given an $[n, k]$ linear binary code c with binary generator matrix $g \in m_{n\times k}$, there exists a matrix $p \in m_{(n-k)\times n}$ such that for any vector s, $gs \in c$ if and only if $ps = 0$. More generally, a vector $v \in \mathbb{F}_2^n$ is a codeword if and only if $pv = 0$.*

Proof. By definition, a linear code c is a k-dimensional subspace of $\mathbb{F}_2^n$. Therefore, we can consider its orthogonal complement $c^\perp \subseteq \mathbb{F}_2^n$, defined as

$$c^\perp = \{w \in \mathbb{F}_2^n \mid \forall v \in c, \langle w, v\rangle = 0\}\,. \tag{5.1}$$

It is known that the orthogonal complement has dimension $\dim(c^\perp) = \dim(\mathbb{F}_2^n) - \dim(c) = n - k$, so $c^\perp$ has a basis $\{p_1, p_2, \ldots, p_{n-k}\}$. We can then construct the matrix

$$p = \begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_{n-k} \end{pmatrix}. \tag{5.2}$$

- ($\Longrightarrow$) Let $v \in c$. By the definition of $c^{\perp}$, we know that for each row p_i, $p_i v = 0$, and therefore $pv = 0$.
- ($\Longleftarrow$) Let $v \in \mathbb{F}_2^n$ such that $pv = 0$. This means that for every row $p_i \in p$, $p_i \cdot v = 0$, implying that $v \in (c^{\perp})^{\perp}$. By Lemma 5.3, we conclude that $v \in c$.

□

Lemma 5.3. *Given a subspace $c \subseteq \mathbb{F}_2^n$ and its orthogonal complement $c^{\perp}$,*

$$(c^{\perp})^{\perp} = c. \tag{5.3}$$

By the definition of $c^{\perp}$, we know that $\forall v \in c$, $v \in (c^{\perp})^{\perp}$. Moreover, by calculating dimensions, we observe that

$$\dim((c^{\perp})^{\perp}) = \dim(\mathbb{F}_2^n) - \dim(c^{\perp}) = n - (n - \dim(c)) = \dim(c). \tag{5.4}$$

Since $c \subseteq (c^{\perp})^{\perp}$ and both have the same dimension, they are the same subspace.

Example 5.4. For example, the repetition code can be thought of as a linear code with generator matrix

$$g = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}. \tag{5.5}$$

It then receives any single bit of information (represented in this case as a vector, as noted earlier), producing:

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}; \quad \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}. \tag{5.6}$$

We can define the parity matrix as

$$p = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}. \tag{5.7}$$

Errors can be checked as follows:

- If $ps = (0, 0)$, there was no error.
- If $ps = (1, 0)$, the error occurred in the second bit.
- If $ps = (0, 1)$, the error occurred in the third bit.
- If $ps = (1, 1)$, the error occurred in the first bit.

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